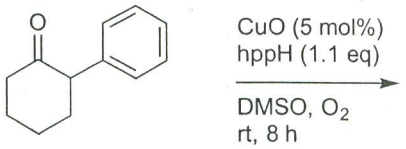
Organic Division Cumulative Exam 12/02/16

Topic: Schoenebeck et al, J. Am. Chem. Soc., 2015, 138, 518-526.

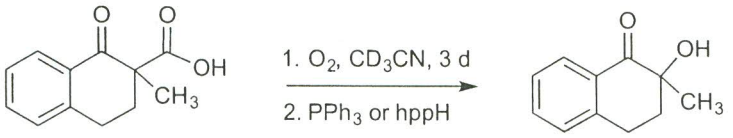
1. Draw the two principle products of the reaction shown that are the focus of the paper.



2. What is the structure of hppH?

3. What is the proposed common intermediate that leads to the two products in Question 1?

4. In control experiments to determine the common intermediate, the beta-ketoacid below was converted into the alpha-hydroxy ketone. A spontaneous reaction of the beta-ketoacid takes place upon dissolution in CD_3CN *before the substrate reacts with oxygen.* Draw the mechanism and product of that reaction step.



5. From the common intermediate, what did the authors conclude were the mechanisms of formation of the two products? Propose electron pushing descriptions of the mechanisms.